SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

DOVUN

Project / Solution Name

AIVANCIN

Name (Leader)

Jad Fahmi

Email & Phone (Leader)

jadfahmi07@gmail.com (phone given in the sdchack.my submission)

University Affiliation(s)

German-Malaysian Institute

Live Prototype URL

https://aivancin--jadfahmi.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/AIVANCIN/aivancin

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AIVANCIN is one Python web service; it calls no external API or model while running. The browser loads a plain HTML, CSS and JavaScript interface from a FastAPI app on Uvicorn, deployed on Replit.

The user's own description goes to /api/suggest. The server splits it into sentences, applies a guard for negated, planned or other people's work, and encodes each sentence with our fine-tuned all-MiniLM-L6-v2 encoder (quantized ONNX on the CPU, ONNX Runtime). Sentence vectors are compared with precomputed vectors for O*NET's 2,087 detailed work activities and 17,578 task statements; the closest activities return with the sentence that triggered them. The user ticks what is true and /api/profile recomputes and validates it.

Deterministic modules then rank roles by activity overlap and build up to three routes, a first learning block from a curated, link-checked course catalogue, AI-usage evidence and scenarios, job outlook, DOSM pay steps and SDG tags (/api/routes, /api/update, /api/cohort). Data are bundled, dated, hashed snapshots (CSV, JSON, NumPy); no database. Answers stay in the browser. A RunPod GPU pod trained the encoder offline once. Diagram: Appendix A.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12 (Replit) / 3.13; HTML, CSS, JavaScript	Server, data builds and tests; interface with no framework or build step	PSF License; open web standards
FastAPI 0.150.0; Uvicorn 0.54.0	Web API and ASGI server	MIT; BSD-3-Clause
ONNX Runtime 1.31.0; Tokenizers 0.23.3; NumPy 2.5.4	Run the encoder on the CPU, tokenise text, vector search over bundled activity vectors	MIT; Apache 2.0; BSD-3-Clause (bundled 0BSD, MIT, Zlib, CC0 parts)
pytest 9.1.1; HTTPX 0.28.1; Playwright with Microsoft Edge; Node.js test runner	Offline tests and scripted browser checks; demo video capture. Not shipped in the app	MIT; BSD-3-Clause; Apache 2.0; MIT
PyTorch 2.8 (pod image); Sentence Transformers 6.1.0; Transformers 5.20.2; Datasets 5.1.0; Accelerate 1.15.0; ONNX 1.23.2	Fine-tuning and ONNX export on the RunPod pod only; not app dependencies	BSD-3-Clause; Apache 2.0 (others)
Jost font; KL Pro font; Pillow; FFmpeg	Wordmark (Jost); interface type (KL Pro, team-owned, pre-existing); artwork resizing; demo video encoding	OFL 1.1; all rights reserved (own licence); HPND; GPL build

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None at run time	The app makes no external API or model call; all data and the model are bundled	Not applicable
RunPod REST and GraphQL API (RunPod)	Development only: create, stop and delete the one training pod; check pods and balance	Sponsor-provided credit
Hugging Face Hub (Hugging Face)	Build time: download the pinned base model and Anthropic Economic Index releases; source watch lists release folders	Free, public
ESCO API (European Commission)	Build time: map O*NET occupations to ISCO and MASCO groups	Free, public
O*NET Center, O*NET OnLine, DOSM storage, NCES	Build time and source watch: download dated public files; read outlook pages	Free, public
Microsoft Learn and Saylor course pages	Build time: course catalogue metadata and link checks; users open links only by clicking	Free, public
Google Fonts, analytics, maps, payments, messaging	None used	Not applicable

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live prototype judges use: https://aivancin--jadfahmi.replit.app. Replit Agent set the run settings	Team lead's Replit account; Cloud Run deployment target
RunPod Pods (RunPod)	One GPU training run: pod 82gxhhkg2yedbz, RTX 4090, about 28 min, US$0.42 (total RunPod use US$0.43)	On-demand, Secure Cloud, US; SDC credit
GitHub (GitHub)	Source code, history and README: https://github.com/AIVANCIN/aivancin	Free organisation repository
Hugging Face Hub (Hugging Face)	Source of the pinned base model revision and the AEI releases (downloads only)	Free, public
Vercel (Vercel)	Static visual documentation at https://aivancin.vercel.app, with current product screenshots; separate from the app	Documentation hosting
Database, storage, authentication services	None: bundled files only; no accounts	Not applicable

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2 (rev. 1110a243), team fine-tune, quantized ONNX	sentence-transformers, Hugging Face	Pre-trained, then fine-tuned once on RunPod; runs on the CPU in the app	Apache 2.0
The same encoder, at build time	As above	Suggests which activities each catalogue course covers; a curator confirms every link before it is published	Apache 2.0
No generative or large language model	Not applicable	The app generates no text: explanations come from fixed templates over computed facts	Not applicable

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0 Database	onetcenter.org/database.html	Public	CC BY 4.0	N
O*NET OnLine job outlook (BLS 2025-2035 projections, Bright Outlook)	onetonline.org	Public	CC BY 4.0	N
Anthropic Economic Index, releases Sep 2025, Jan, Mar and Jun 2026	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY (per release)	N
DOSM lfs_qtr_sru_age, lfs_qtr_sru_sex, wages_occupation	open.dosm.gov.my	Public	CC BY 4.0	N
ESCO O*NET-ISCO crosswalk	esco.ec.europa.eu	Public	EC reuse, Decision 2011/833/EU	N
NCES CIP 2020 to SOC 2018 crosswalk	nces.ed.gov/ipeds/cipcode	Public	US public domain	N
Course catalogue: 90 Microsoft Learn and Saylor entries (links, durations, stated objectives)	learn.microsoft.com; saylor.org	Public (curated metadata)	Microsoft terms; Saylor CC BY 3.0	N
Personas, test sets, held-out set and 60-person cohort	Team-written (data/, tests/acceptance/)	Synthetic	Apache 2.0 (project)	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic; Claude Opus 5.5 and Sonnet 5.5)	Code, tests, data tools, README, rendered checks, submission documents and demo video assembly	Major
Codex (OpenAI; GPT-6 family)	Project-selection research, early interface work, proposal/deck drafts and final documentation/submission refresh	Moderate
ChatGPT image generation (OpenAI GPT image)	Welcome artwork (ribbon mountain illustration). The app icon was designed by Jad Fahmi	Minor
Antigravity (Google, Gemini); Replit Agent (Replit)	Documentation website work and one design review; Replit run settings and publishing the deployment	Minor; minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Jad Fahmi set every task, reviewed the results and made the scope, design and data decisions. AI-assisted code entered the repository only after it passed the offline test suite (354 pytest tests) and scripted Edge checks at phone and desktop sizes. Acceptance test sets were written and committed before their first run, and failed checks are reported in the README, not hidden. Every figure the app shows is traced to a dated, bundled source. In the product, model output is only a suggestion the user must confirm. The team takes responsibility for all submitted material.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	US$25 GPU credit (team account)	One fine-tuning pod (RTX 4090, about 28 min); US$0.43 of the credit used in total
Replit	Platform (deployment; Replit Core for participants)	Hosts the live prototype: https://aivancin--jadfahmi.replit.app
None other	No sponsor GenAI API, data or mentoring used	Not applicable

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Neither is a hard runtime dependency, but both improved the delivered prototype. Replit hosts the live deployment judges use; the app has no Replit-specific code and runs the same way locally or on any small Python host, so hosting could be replaced. RunPod supplied the GPU for one fine-tuning run that improved activity suggestions (45 against 39 of 60 held-out descriptions, gate fixed before training). The tuned model is bundled and runs on the CPU, so the app never calls RunPod; had the run failed its gate, the base model would have shipped.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State “No user data is collected” if applicable.

[Maximum 150 words]

No account or user profile is created and the app sets no cookies or analytics. A user's description is sent only to the AIVANCIN server, processed in memory for that request and not stored or logged by the app. No text goes to a third-party service or AI API: the model runs on our own server. Answers, confirmed activities, the chosen route and ticked steps are kept in the user's own browser (local storage) so they can continue later; Clear my data in Profile deletes them. A self-introduction such as "My name is ..." is not read. Demo people, test sets and the career-centre cohort are synthetic; no real CVs or personal data were collected or used. The app needs no API keys or secrets. The hosting platform may keep its own standard access logs under its terms.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Suggestions come from a small English sentence encoder: it can miss idioms and short forms and does not read Malay, so every suggestion needs the user's tick. A word-list guard holds back negated, planned or other people's work but misses some phrasings. Role data come from US O*NET; Malaysia appears only as observed DOSM group statistics, never projected. AI-usage figures describe global Claude conversations, not Malaysian labour demand. The moderate AI-shift projection has a backtested mean error of 16.3 points and uncalibrated ranges, so values are labelled rough estimates. On synthetic profiles, the newest predeclared set has complete routes for 7 of 12 in-scope profiles, below our 80% target, with missed work and off-family routes even within supported fields. Mitigations: user confirmation, sources and dates on every figure, explicit "no verified resource" and exploration states, keyword fallback if the model fails, and no generated text. One host, no redundancy.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [ ] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

Items 6 and 7 are left for the team leader to tick once the repository is public and this declaration is signed.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

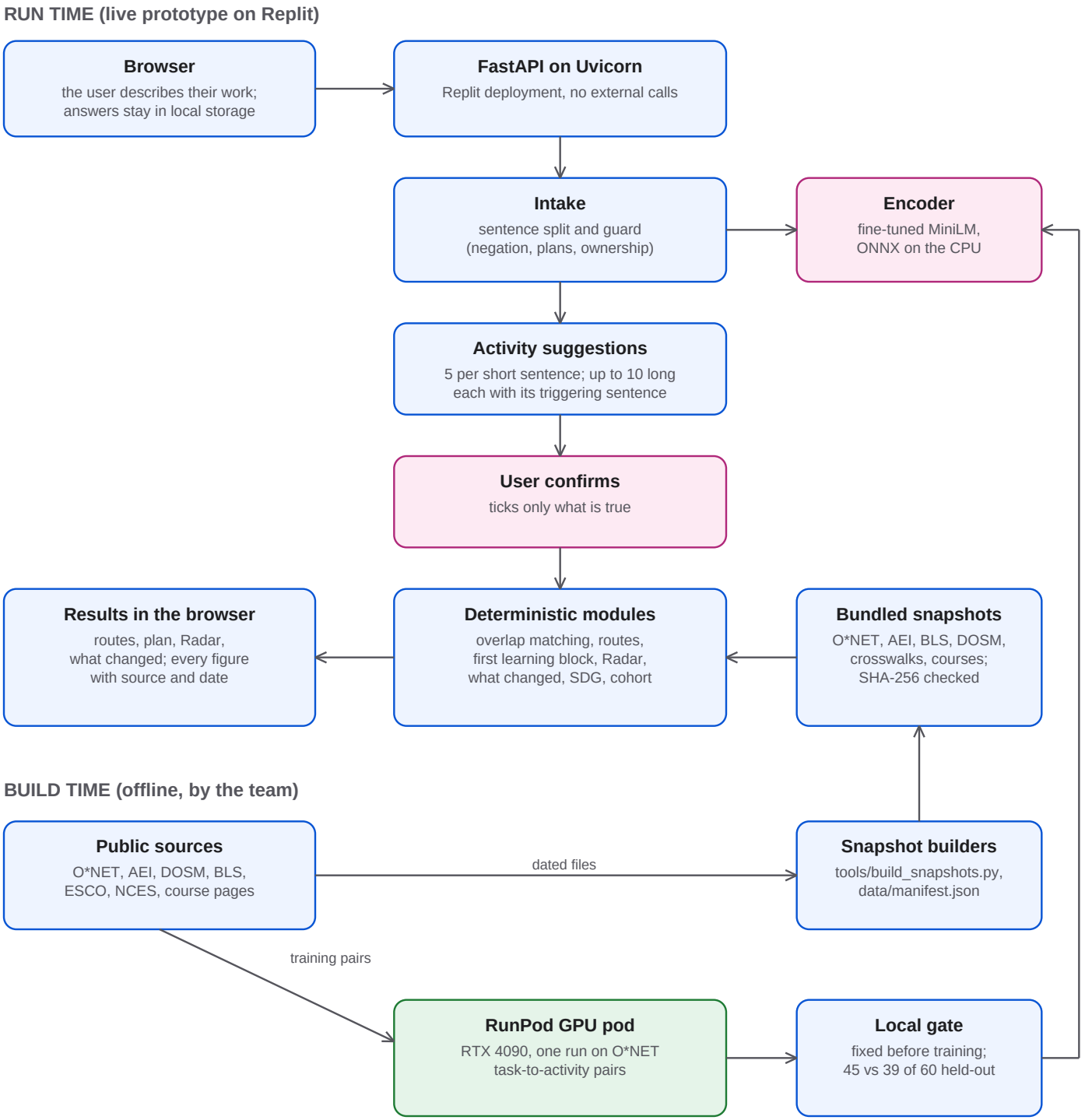
Jad Fahmi

[Date]

25 October 2026

Appendix A: System block diagram

DOVUN | AIVANCIN | Technical and Tools Declaration | repository copy

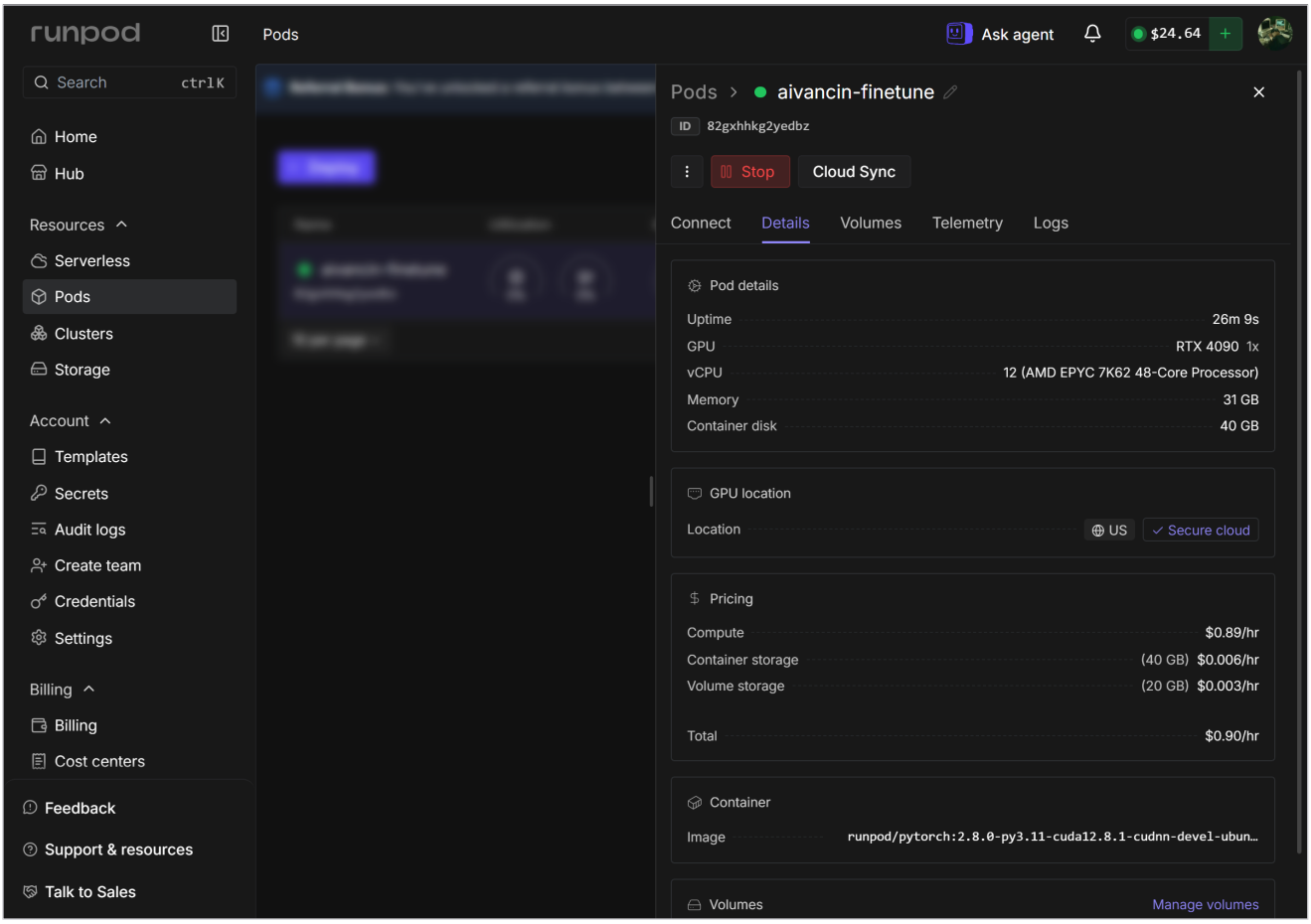


The app never calls RunPod or any AI API. A missing or altered model file switches the app to keyword search with a clear message; an altered dataset stops start-up.

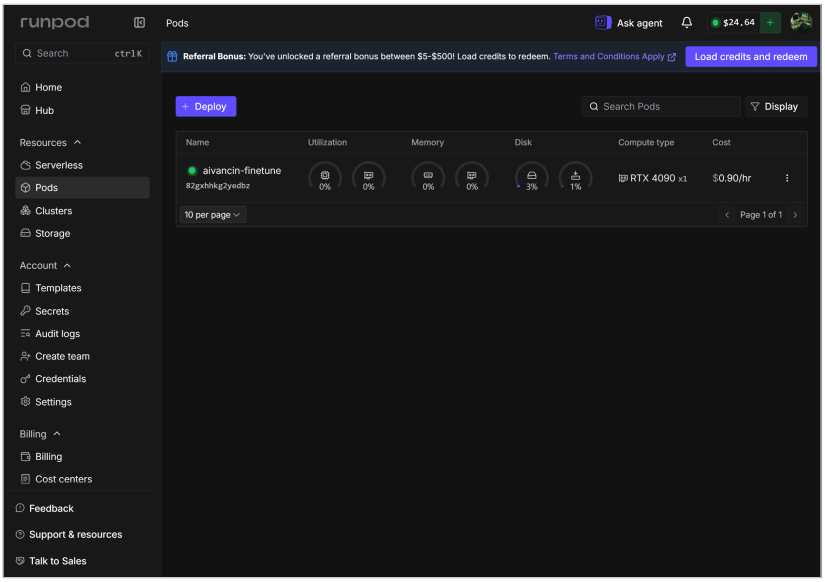
Appendix B: RunPod proof of use

Dashboard screenshots from the team account, 21 October 2026 (PH 10.7)

Pod **82gxhhkg2yedbz** ("aivancin-finetune"), NVIDIA GeForce **RTX 4090** x1, on-demand Secure Cloud (US), image runpod/pytorch:2.8.0-py3.11-cuda12.8.1-cudnn-devel-ubuntu22.04. Pod started 21 Oct 2026 18:41 MYT and stopped 19:09:55 MYT (about 28 min; uptime 26m 9s when captured). Training took 15.6 s, 25.5 s including exports; cost US$0.42, total RunPod use US$0.43 of the US$25 credit. The pod was then deleted. Record in the repository: data/model/finetune.json.



Pod details: ID, GPU, uptime and pricing.

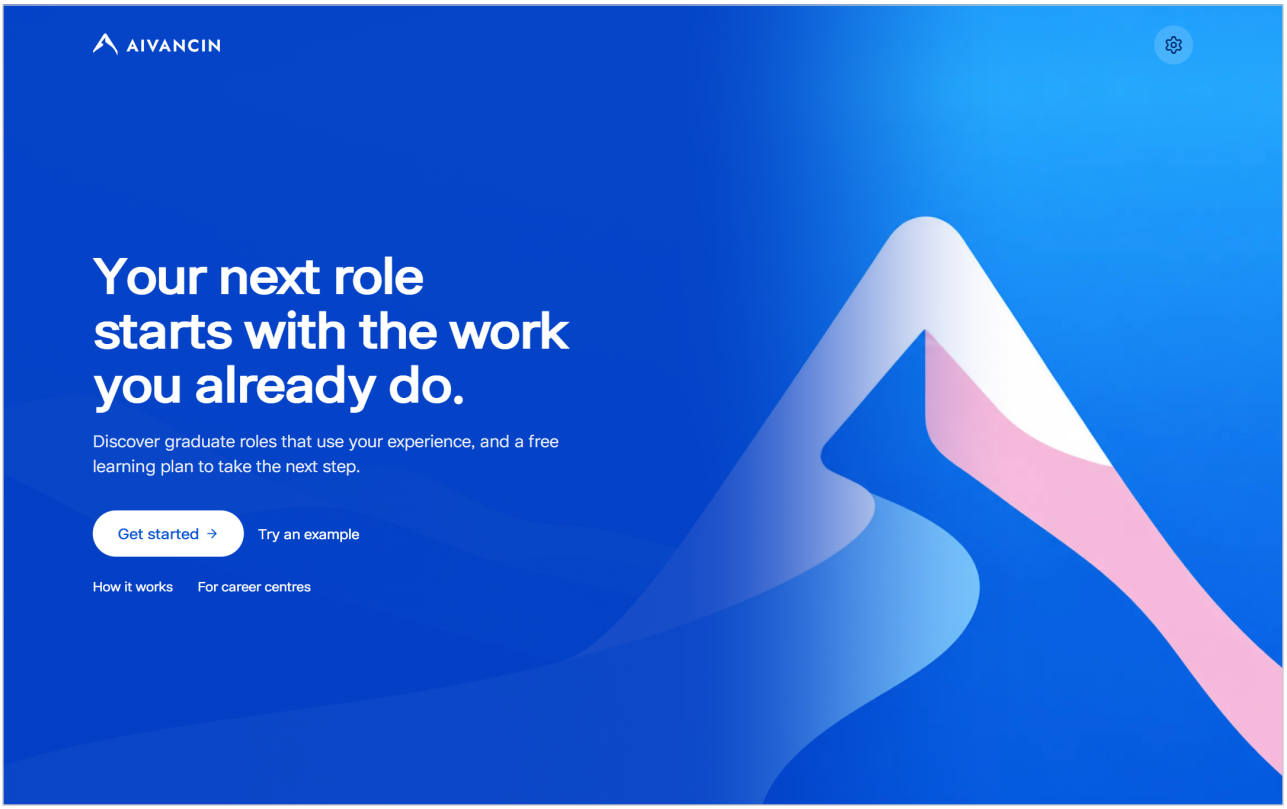


Pods list: the same pod running on the team account.

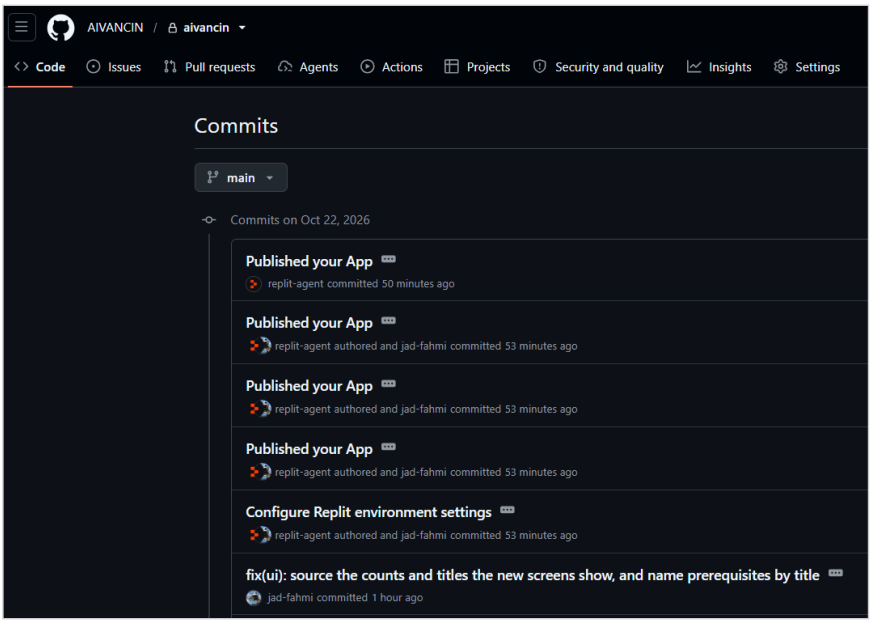
Appendix C: Replit proof of use

Live deployment on the team lead's Replit account (PH 10.6)

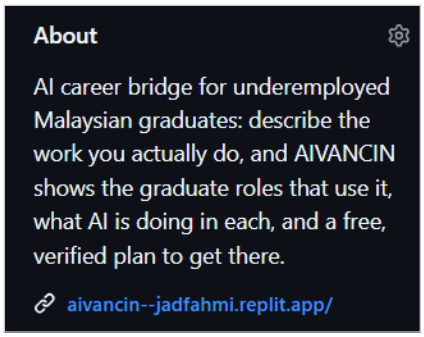
Live deployment URL: **https://aivancin--jadfahmi.replit.app** (Replit account "jadfahmi", the team leader). Checked 2026-10-23 14:14 MYT: HTTP 200, AI suggestions available; the served app.js and app.css are byte-identical to the repository's main branch. Replit Agent configured the run settings and published the app; its commits are in the repository history.



The live prototype at the URL above.



Repository history: Replit Agent commits "Configure Replit environment settings" and "Published your App".



Repository About links the live URL.